Revision sheet for plate tectonics

Answer the following questions with as much detail as possible. When you have finished, check your answers with the person beside you. Correct answers will be provided at the end of the lesson.

1. Draw a labelled diagram of the internal structure of the Earth.
2. List some physical differences in the properties of the Earth's crust compared to the Earth's core.
3. Draw a diagram showing an example of folding in rocks which causes faulting and label the forces with arrows.
4. What is the difference between the epicentre and focus of an earthquake?
5. Earthquakes have caused large or small cracks in the base of the memorial statues shown below.



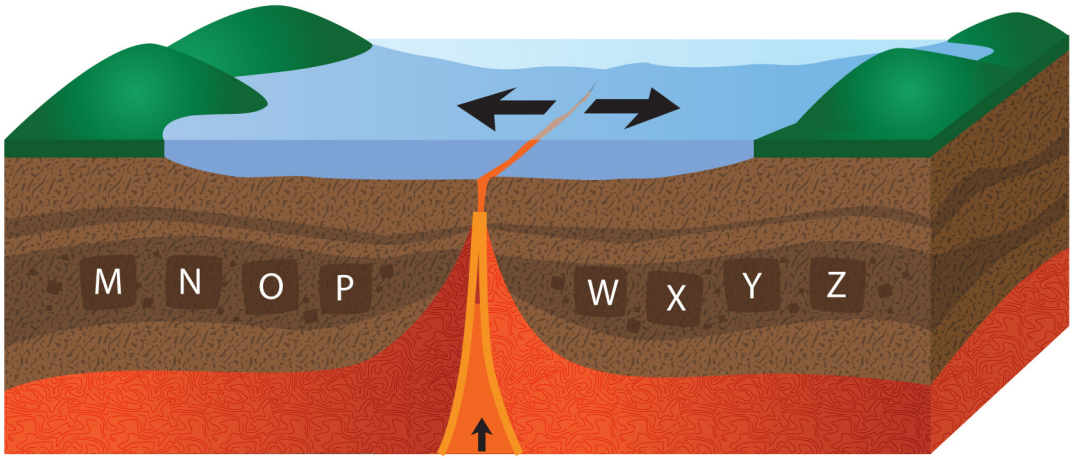
The memorial statues show the year the soldier was born and when the soldier died. Use the dates on the statue to choose the correct conclusion.

a) No earthquake occurred between 1865 and 1938
b) A small earthquake occurred before 1938
c) A large earthquake occurred between 1905 and 1910
d) A large earthquake occurred between 1836 and 1892

6. Draw a diagram of a subduction zone. Label the diagram and answer the follow questions.
   a) Where are earthquakes most likely to occur?
   b) Where are earthquakes least likely to occur?
   c) What type of plate boundary is this?
   d) What two types of crust do you need for a subduction zone?
   e) What happens when two continental crusts converge?

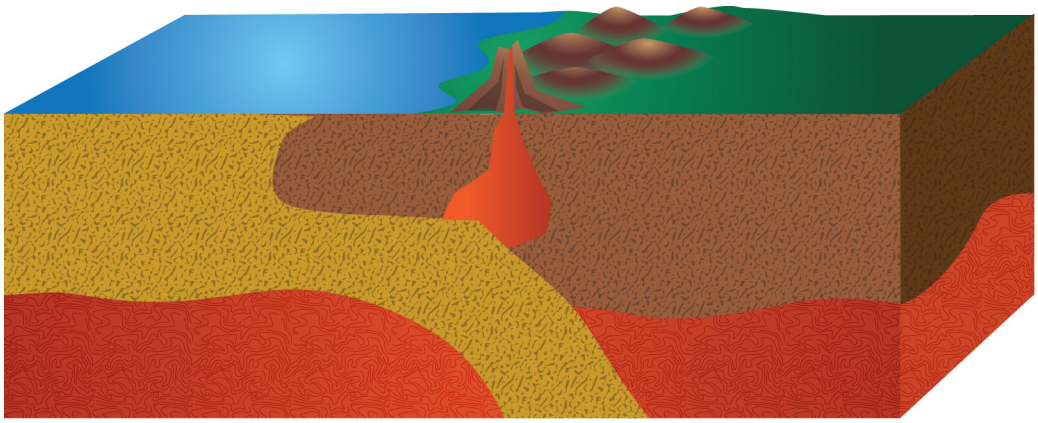
7. Explain why earthquakes and volcanoes are more likely to be found at the edges of plates.
8. It is believed that Australia was once part of a larger land mass called Gondwana and connected to South America.
   a) What is the name given to the process that caused Gondwana to split and move apart?
   b) What evidence is there to support the existence of the supercontinent?

9. List at least three different types of evidence that supports the theory of continental drift.
10. Examine the picture below of a mid-ocean ridge and answer the following questions.



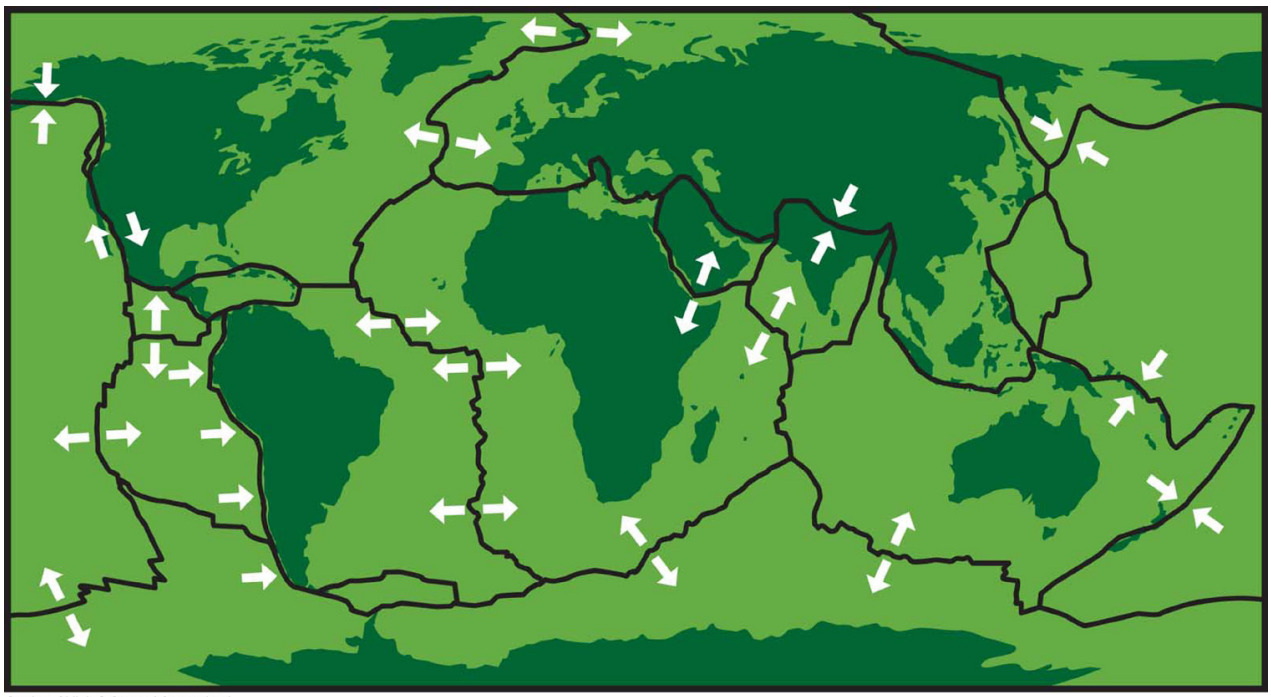
   a) Which rock would be the same age as rock N?
   b) Why have you chosen your answer?

11. The following is a diagram of a subduction zone. Examine the diagram and answer the following questions.



   a) In which directions are plate A and plate B moving?
   b) Give a reason why rock forced downwards into the subduction zone is likely to melt.
   c) What is believed to be the driving force for tectonic action?

12. What is responsible for the formation of new crust at the edge of a tectonic plate?
13. The Earth's crust is divided into a number of tectonic plates that are slowly moving. Look at the diagram below and answer the questions.



Gordon, Shiloh © State of Queensland

a) Which plates are sliding past each other?
b) Which plates are colliding?
c) Which plates are moving away from each other?
d) Which boundaries are likely to experience earthquakes but not volcanoes?

14. A rift valley is evidence of which kind of plate boundary?
15. What will a rift valley eventually become?
16. The convergence of two continental plates would produce which type of landform?